Energy Conservation In a Skate Park

Synopsis: In this note, you will utilize the PhET simulation to observe the skater's kinetic (KE) and potential energy (PE) as the skater skates on a ramp. Are the changes in KE and PE related? If so, how?

First, simulate the ramp with no friction. Then, turn on the friction and see how it affects KE, PE, and the total energy.

Performance Expectations: Describe energy conversion and conservation of the skater's energy on a half-pipe.

For PHY 151 — with calculus: Treat the constant total energy as an equation and differentiate it in time; recover Newton's second law along the track.

Instructions and Questions

1) Type "Skate Park PhET" into the Google search bar and find Energy Skate Park - PhET.
2) Open the Energy Skate Park simulation, and choose Intro.
3) Click the plus sign next to the Energy button on the left to display the KE, PE, Thermal, and Total energy levels bar chart.
4) Check the "Pie Chart," "Speed," "Path," and "Stick to Track" boxes on the options panel on the right.
5) Place the skater at the top of the half-pipe.
   Q1) What happens to the skater?

   Q2) What happens to the PE, KE, Thermal, and total energy?

6) Sketch the bar chart below. Pause the motion at the top and bottom.

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Top Right of Ramp	Bottom of Ramp	Top Left of Ramp
		
		

7) Discuss the energy changes that you see in the bar charts. The bar chart shows the following energies:
   - Energy of motion (Kinetic energy)
   - Energy due to the skater's height above ground (Gravitational potential energy)
   - Total mechanical energy (Kinetic energy + Gravitational Potential Energy)
   - Thermal Energy (Heat energy generated by friction)

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Q3) Answer the following questions:

i) **Which energy bar stays constant ALL THE TIME:**

ii) **Which energy is at a maximum when the skater is at the top of the pipe:**

iii) **Which energy is at a maximum when the skater is at the bottom of the pipe:**

iv) **Which energy always stayed at zero? (This makes sense because there is no friction.)**

8) Looking at the speed dial (check the Speed box in the option panel on the right), discuss the skater's speed at the top and bottom of the ramp.

Q4) Answer the following questions:

i) **What is the speed at the top?**

ii) **What is the speed at the bottom?**

iii) **How are the speeds related to KE and PE at the top and bottom of the ramp?**

9) Turn on the friction by sliding the friction from "None" to a mid-range value.

10) Sketch the bar chart below. Pause the motion at the top and bottom.

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Top Right of Ramp	Bottom of Ramp	Top Left of Ramp
		
		

Q5) What happens to the skater over time with friction turned on?

Q6) Why does the skater eventually stop?

11) Summarizing, answer the following questions.

Q7) What happens to the KE and PE as the skater rolls down the hill (Hint: discuss the energy conversion)

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Q8) If there is no friction, what can we conclude about the total mechanical (PE+KE, NOT including thermal) energy?

Q9) What kind of energy is generated when friction is present?

Q10) How does friction affect the total mechanical energy of the skater?

12) Bonus. Set the friction back to "None", check "Reference Height" on the options panel, and drag the dashed reference line up until it sits above the bottom of the pipe.
Q11) What happens to the PE bar when the skater is below the reference line? What does a negative potential energy mean? Did the skater's motion change at all when you moved the line?

The conservation statement is an equation you can differentiate.
With no friction the bar chart says $E = \frac{1}{2}mv^2 + mgh$ = constant along the whole ride. A constant has zero time derivative, so $d/dt(\frac{1}{2}mv^2 + mgh) = 0$. Carry out the derivative with the chain rule and you get $mv\cdot(dv/dt) + mg\cdot(dh/dt) = 0$. Along the track, $dh/dt = v\cdot(dh/ds)$ where s is distance along the track, so after dividing by mv the statement becomes $dv/dt = -g\cdot(dh/ds)$: the acceleration along the track is minus g times the local slope. That is Newton's second law for the skater, and it came out of "total energy is constant" without drawing a single force. It is also the $F = -dU/dx$ of Note 1 in disguise.

i) At the bottom of the pipe the speed is a maximum. What is dv/dt there, and why does that agree with the slope of the track at that point?

ii) Where on the half-pipe is $|dv/dt|$ largest? Is that where the skater is fastest? Explain the difference between "fastest" and "accelerating hardest" using the slope.

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